

Functional Annotation of *Caenorhabditis elegans* NDX-4 (Ap4A hydrolase, UniProt Q9U2M7)

Summary

NDX-4 (gene *ndx-4*; ORF Y37H9A.6; UniProt Q9U2M7) is the *Caenorhabditis elegans* **asymmetrical diadenosine 5',5'''-P¹,P⁴-tetrphosphate (Ap4A) hydrolase**, a small metal-dependent enzyme of the **Nudix hydrolase superfamily** (EC 3.6.1.17). Its primary, biochemically defined function is to **cleave the metabolite Ap4A asymmetrically at the fourth phosphate to yield ATP + AMP**. The enzyme acts on a range of related **dinucleoside tetrphosphates (Ap4N) but not on dinucleoside triphosphates (Ap3A)**, a specificity dictated by precise orientation of the substrate in the active site. The worm enzyme is historically important: it was the first eukaryotic asymmetrical Ap4A hydrolase whose crystal structure was solved, and that structure is the defining prototype for the entire asymmetrical Ap4A-hydrolase subfamily, used as the reference model for human, bacterial, and other orthologues.

Mechanistically, NDX-4 uses the canonical **mixed α/β Nudix fold** with a catalytic Nudix motif. Site-directed mutagenesis of the worm enzyme established the catalytic machinery in quantitative detail: wild-type $k_{cat} \approx 23 \text{ s}^{-1}$ and $K_m \approx 8.8 \text{ }\mu\text{M}$ for Ap4A, with three conserved glutamates (Glu56, Glu52, Glu103) serving as the P₄-phosphate-binding catalytic residues and His31/Lys36/Lys83 stabilising the P₁-phosphate of the substrate. The same single active site also carries a **minor, secondary PRPP-pyrophosphatase activity**, converting 5-phosphoribosyl-1-pyrophosphate (PRPP) to ribose-1,5-bisphosphate + Pi — but this is a low-efficiency side reaction, not the enzyme's principal physiological job.

Biologically, NDX-4 is the **catabolic "off-switch" of the Ap4A stress-signalling system**. Ap4A is a universal, stress-induced dinucleoside polyphosphate (an "alarmone") produced largely as a side-product of aminoacyl-tRNA synthetases — classically lysyl-tRNA synthetase (LysRS). Ap4A concentration is the key variable: at controlled levels it can act as an intracellular second messenger influencing proliferation, quiescence, differentiation, and apoptosis ("friend"), whereas uncontrolled accumulation is toxic ("foe"), disturbing zinc homeostasis and spuriously

occupying ATP-binding sites. By degrading Ap4A, NDX-4 keeps intracellular Ap4A low and sets the Ap3A:Ap4A ratio, thereby governing whether this nucleotide behaves as signal or toxin. The enzyme is a **soluble intracellular protein with a probable nuclear site of action**, inferred from the nuclear localisation of its closely related orthologues.

Gene / Protein Identity Verification

Before presenting findings, the identity of the research target was confirmed against the UniProt record and the primary literature. All verification criteria are satisfied:

Criterion	UniProt record	Literature match
Gene symbol	<i>ndx-4</i> (Nudix hydrolase 4)	Matches ORF Y37H9A.6, the <i>C. elegans</i> Ap4A hydrolase
Organism	<i>Caenorhabditis elegans</i>	Crystal structures and mutagenesis performed on the <i>C. elegans</i> enzyme (PMID: 11937063 , PMID: 12475970)
Protein family	Nudix hydrolase family	Structure reveals "the mixed α/β fold of the Nudix family" (PMID: 11937063)
Function	Asymmetrical Ap4A hydrolase (EC 3.6.1.17)	Asymmetric cleavage of Ap4A $\rightarrow$ ATP + AMP, directly demonstrated
Domains	NUDIX hydrolase domain; Tetra_PHTase	Consistent with asymmetrical Ap4A-hydrolase subfamily

Conclusion: The gene symbol is *not* ambiguous for this target. There is a direct, high-quality primary-literature match — the *C. elegans* enzyme is one of the best structurally and biochemically characterised Ap4A hydrolases in any organism. The research below pertains unambiguously to Q9U2M7 / NDX-4.

Key Findings

Finding 1 — NDX-4 is the *C. elegans* asymmetrical Ap4A hydrolase, a Nudix-fold enzyme

The crystal structure of the *C. elegans* Ap4A hydrolase was determined for both the **free enzyme (2.0 Å)** and a **binary complex (1.8 Å)**. These structures revealed that the enzyme adopts "the mixed alpha/beta fold of the Nudix family," placing NDX-4 firmly within the Nudix hydrolase superfamily and identifying its catalytic Nudix motif ([PMID: 11937063](#)). Functionally, the enzyme (EC 3.6.1.17) **asymmetrically cleaves** diadenosine 5',5'''-P¹,P⁴-tetraphosphate (Ap4A) into **ATP + AMP** — "asymmetrical" meaning the phosphoanhydride bond cleaved lies at the fourth phosphate counting from one adenosine, so the two products are unequal (ATP vs AMP) rather than two identical ADP molecules (which would be symmetrical cleavage, EC 3.6.1.41). This finding establishes the fundamental catalytic identity of the gene product.

Finding 2 — The catalytic machinery is defined in quantitative detail by site-directed mutagenesis

A systematic mutagenesis study replaced 13 residues predicted from the enzyme-inhibitor crystal complex and measured the kinetic consequences for the worm enzyme. The **wild-type enzyme has $k_{cat} = 23 \text{ s}^{-1}$ and $K_m = 8.8 \text{ }\mu\text{M}$ for Ap4A** ([PMID: 12475970](#)). The key catalytic residues are three conserved glutamates that bind the P4 phosphate: replacing **Glu56, Glu52, and Glu103 with glutamine reduced k_{cat} by 10^5 -, 10^3 -, and 30-fold respectively** — the hallmark of the conserved Nudix catalytic glutamates that coordinate the divalent metal and activate the attacking water. Separately, substrate-binding residues that stabilise the P1 phosphate were identified: "mutating His31 to Val or Ala and Lys83 to Met produced 10- and 16-fold increases in K_m ," and Lys36/Lys83 also contribute to catalysis. Together these data provide a residue-by-residue map of how NDX-4 binds and hydrolyses its substrate.

Residue	Role	Effect of mutation
Glu56	Catalytic (P4-phosphate / metal)	kcat ↓ 10 ⁵ -fold
Glu52	Catalytic (P4-phosphate / metal)	kcat ↓ 10 ³ -fold
Glu103	Catalytic (P4-phosphate / metal)	kcat ↓ 30-fold
His31	P1-phosphate binding	Km ↑ 10-fold
Lys83	P1-phosphate binding + catalysis	Km ↑ 16-fold
Lys36	P1-phosphate binding + catalysis	Km increase

Finding 3 — Substrate specificity: dinucleoside tetraphosphates, not triphosphates

Structural analysis showed how NDX-4 "can catalyze the hydrolysis of a range of related dinucleoside tetraphosphate, but not triphosphate, compounds through precise orientation of key elements of the substrate" (PMID: 11937063). In other words, the enzyme accepts a family of Ap4N substrates (varying the second nucleoside) provided there are four bridging phosphates, but it discriminates against Ap3A (three phosphates). This specificity is central to its physiological role: by preferentially acting on tetraphosphates, the enzyme "has a key role in regulating the intracellular Ap4A levels and hence potentially the cellular response to metabolic stress and/or differentiation and apoptosis via the Ap3A/Ap4A ratio" (PMID: 11937063). The orthologous chemistry is confirmed in the *Drosophila* enzyme, where "diadenosine tetraphosphate is hydrolysed to ATP and AMP" (PMID: 17344088), with Km ~9–12 μM and kcat 13–43 s⁻¹ — kinetics closely matching the worm enzyme.

Finding 4 — Ap4A hydrolases act (in orthologues) in the nucleus and regulate a stress-induced second messenger

Subcellular-localisation data from the closely related *Drosophila* orthologue place the enzyme in the nucleus: "Apf-EGFP ... reveals Apf to be predominantly nuclear, having an apparent preferential association with euchromatin and facultative heterochromatin" (PMID: 17344088). This supports a nuclear site of action for Ap4A metabolism and, by inference, for NDX-4 (which lacks direct *C. elegans* localisation data). The regulatory importance of this degradation is shown in mammalian cells, where "the 'Nudix' type 2 gene product, Ap4A hydrolase, is responsible for Ap4A degradation following the immunological activation of mast cells" (PMID: 18644867); knockdown raises Ap4A and alters MITF/USF2 target-gene expression, helping

establish Ap4A as a second messenger for gene regulation. Ap4A and related dinucleoside polyphosphates accumulate 30–70-fold under oxidative/metabolic stress, acting as "alarmones."

Finding 5 — A secondary PRPP-pyrophosphatase activity using the *same* active site

In a survey of 17 Nudix hydrolases, all 11 enzymes active toward dinucleoside polyphosphates (≥ 4 phosphates) — including the *C. elegans* Ap4A hydrolase — also hydrolysed **5-phosphoribosyl-1-pyrophosphate (PRPP)**, with "the products of hydrolysis ... ribose 1,5-bisphosphate and Pi" (PMID: 12370170). Critically, "active site mutants of the *Caenorhabditis elegans* diadenosine tetraphosphate hydrolase had no activity, confirming that the same active site is responsible for nucleotide and PRPP hydrolysis." This is a clean demonstration of catalytic promiscuity sharing one active site. However, comparison of specificity constants indicated PRPP is only a **minor/secondary substrate** for most of these enzymes — so while mechanistically informative (and potentially linking to the glycolytic activator ribose-1,5-bisphosphate), it should not be mistaken for NDX-4's principal physiological reaction.

Finding 6 — Substrate-binding mode defined via a non-hydrolysable analogue; the worm structure is the subfamily template

The binary complex of the worm enzyme was obtained with the **non-hydrolysable Ap4A analogue AppCH2ppA** (a β,γ -methylene bridge analogue): "Crystals of this enzyme from the nematode *Caenorhabditis elegans* have been obtained in the presence of a non-hydrolysable substrate analogue, AppCH2ppA" (PMID: 11856844). This complex revealed that the adenine moiety binds in a **ring-stacking arrangement**. That geometry has since served as the universal reference for the subfamily: the human enzyme structure notes "the adenine moiety of the nucleotide predominantly binds in a ring stacking arrangement equivalent to that observed in the x-ray structure of the homologue from *Caenorhabditis elegans*" (PMID: 15596429). Orthologue structures from *Aquifex aeolicus* (PMID: 20124691) and *Chlamydia trachomatis* CT771 (PMID: 24354275) likewise mirror the *C. elegans* ligand-bound geometry, underscoring NDX-4's role as the structural prototype of asymmetrical Ap4A hydrolases.

Finding 7 — NDX-4 is the catabolic arm of the Ap4A cycle (LysRS makes it, NDX-4 removes it)

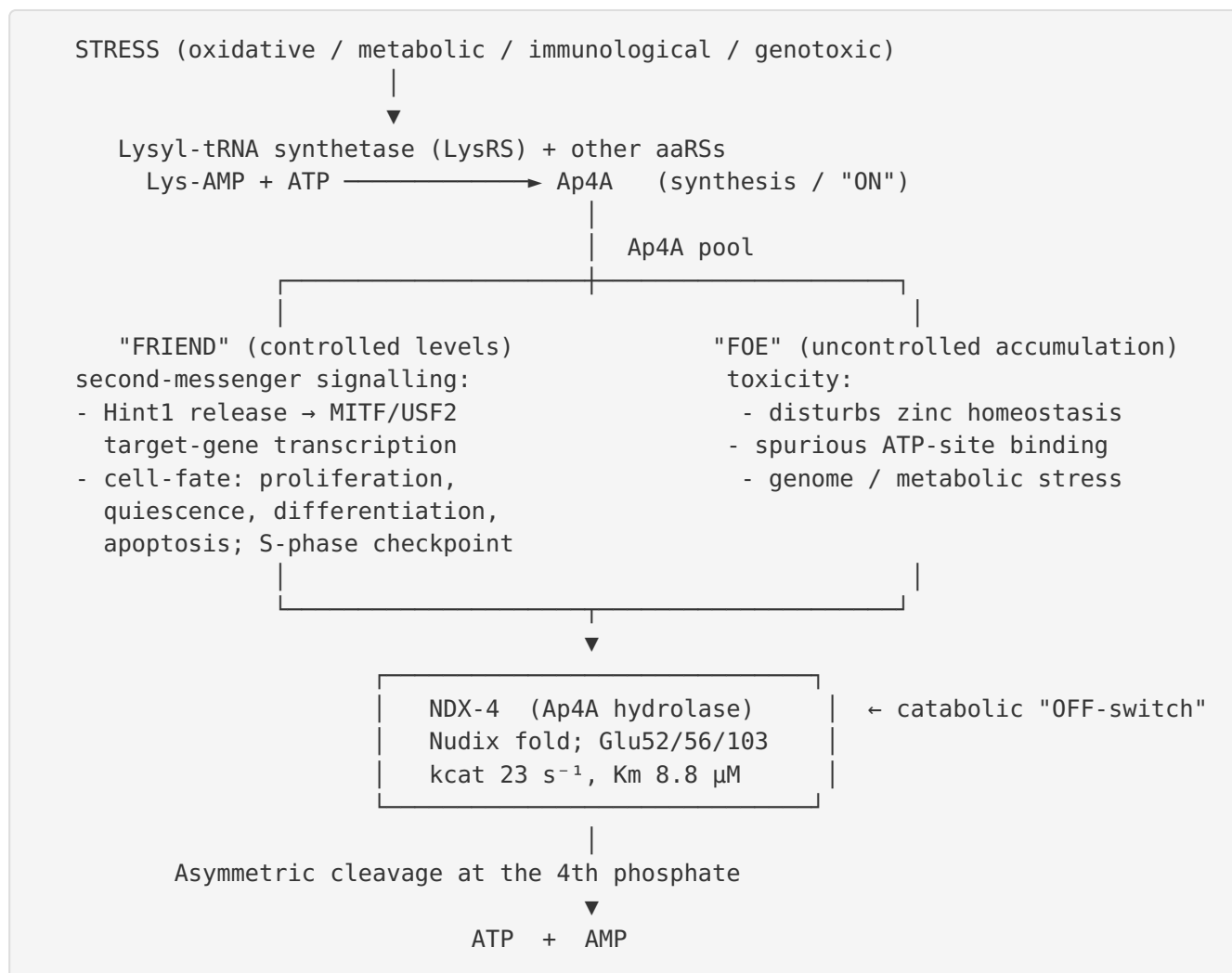
Ap4A is produced as a side-product of aminoacyl-tRNA synthetases, classically **lysyl-tRNA synthetase (LysRS)**, which condenses Lys-AMP with ATP. In the LysRS–Hint1–MITF signalling axis, phosphorylated LysRS translocates to the nucleus where its C-terminal domain "trigger[s] LysRS-directed production of the second messenger Ap4A that activates MITF" (PMID: 23159739). NDX-4 is the opposing catabolic arm. A critical reassessment in *E. coli* argues that Ap4A removal is largely constitutive: "we found that Ap4A is efficiently removed in a constitutive, nonregulated manner," and that "accumulation at very high levels is toxic due to disturbance of zinc homeostasis" (PMID: 28516732). This frames NDX-4 as a housekeeping hydrolase whose job is to keep a potentially damaging metabolite at low levels — whether Ap4A is a *bona fide* second messenger or primarily a damage metabolite, the hydrolase is the control valve.

Finding 8 — Ap4A levels (set by hydrolases like NDX-4) govern cell-fate decisions: a concentration-dependent friend/foe model

Authoritative reviews establish that dinucleoside polyphosphates are produced by aminoacyl-tRNA synthetases and degraded by specific hydrolases, with concentration-dependent roles. McLennan's review notes "recent evidence supporting roles for Ap3A and Ap4A in the cellular decision making processes leading to proliferation, quiescence, differentiation, and apoptosis," and crucially that "the roles of friend and foe are not incompatible, but are distinguished by the concentration range of nucleotide achieved under different circumstances" (PMID: 11007992). A more recent review confirms universality: "In response to various environmental and genotoxic stresses, all cells produce dinucleoside polyphosphates" (PMID: 33282915). NDX-4 is the enzyme that sets where on the friend/foe concentration axis the cell sits.

Mechanistic Model / Interpretation

NDX-4 sits at the catabolic node of a biosynthesis–degradation cycle that controls the intracellular concentration of the stress nucleotide Ap4A:



Catalytic logic. NDX-4 is a metal-dependent phosphohydrolase. The three catalytic glutamates (Glu52/56/103) in the Nudix motif coordinate the divalent cation(s) and activate a water nucleophile that attacks the fourth phosphate; His31/Lys36/Lys83 anchor the P1 phosphate to orient the substrate. Because the active site reads out the *length* of the polyphosphate bridge, Ap4N substrates are accepted but Ap3A is excluded. The adenine ring-stacking pocket (defined by the AppCH2ppA complex) provides nucleobase recognition. The same site can accommodate PRPP, hydrolysing it to ribose-1,5-bisphosphate + Pi, but with much lower efficiency — a textbook example of Nudix catalytic promiscuity.

Physiological logic. The enzyme's purpose is homeostatic: keep Ap4A low and set the Ap3A:Ap4A ratio. Whether one adopts the "second-messenger" interpretation (Ap4A actively signals via Hint1/MITF to control gene expression and cell fate) or the more conservative "damage metabolite" interpretation (Ap4A is a toxic by-product that must be scavenged), NDX-4's role is identical — it is the valve that prevents accumulation. Nuclear localisation of

orthologues suggests the relevant Ap4A pool, and hence NDX-4's site of action, is at least partly nuclear/chromatin-associated, consistent with a role in nucleotide-signal control close to the transcriptional machinery.

Evidence Base

PMID	Study	How it supports the findings
11937063	Crystal structure of <i>C. elegans</i> Ap4A hydrolase (free + binary)	Establishes Nudix fold, asymmetric ATP+AMP chemistry, tetra-not-tri specificity, and regulatory role via Ap3A/Ap4A ratio (F1, F3)
12475970	Site-directed mutagenesis of catalytic/binding residues	Quantitative catalytic map: k_{cat} 23 s ⁻¹ , K_m 8.8 μ M; Glu52/56/103 catalytic, His31/Lys36/Lys83 substrate binding (F2)
12370170	Nudix enzymes with PRPP pyrophosphatase activity	Shows the same worm active site hydrolyses PRPP $\rightarrow$ ribose-1,5-bisP + Pi (F5)
11856844	Crystallisation with AppCH2ppA analogue	Identifies the non-hydrolysable analogue defining substrate binding (F6)
15596429	Human Ap4A hydrolase structure	Confirms worm structure as the adenine-binding reference for orthologues (F6)
20124691	<i>Aquifex aeolicus</i> Ap4A hydrolase (ATP-bound)	Orthologue mirrors worm ring-stacking geometry (F6)
24354275	<i>Chlamydia</i> CT771 (nudH)	Bacterial asymmetric Ap4A hydrolase mirroring the worm structure (F6)
17344088	<i>Drosophila</i> Ap4A hydrolase characterisation	ATP+AMP chemistry; predominantly nuclear localisation (F3, F4)
18644867	Ap4A hydrolase in mast-cell transcription	Degradation of Ap4A second messenger regulates gene expression (F4)
23159739	LysRS structural switch (translation/transcription)	Defines LysRS as the biosynthetic source of Ap4A that NDX-4 counteracts (F7)
19524539	LysRS Ser207 phosphorylation / Ap4A production	Upstream regulation of the Ap4A signal (F7)
22329685	Hint1 adenylate recognition	Downstream Ap4A effector in signalling (F7)
28516732	Ap4A: alarmone or damage metabolite? (<i>E. coli</i>)	Shows constitutive Ap4A removal; toxicity via zinc homeostasis (F7)
11007992	"Dinucleoside polyphosphates — friend or foe?" review	Concentration-dependent friend/foe model of cell-fate control (F8)

PMID	Study	How it supports the findings
33282915	Re-evaluation of Ap4A review	Universality of stress-induced dinucleoside polyphosphate production (F8)
30700216	House-cleaning enzymes in <i>Plasmodium</i>	Ap4A hydrolase essential in a parasite — underscores physiological importance of the activity (context)
23184251	Nudix substrate ambiguity review	Frames the PRPP/antimutator promiscuity and annotation caution (context for F5)

Convergence and tension in the evidence. The structural and enzymological evidence (F1–F3, F6) is direct, high-quality, and specific to the *C. elegans* enzyme — unusually strong primary data for a non-model-organism gene. The signalling interpretation (F4, F7, F8) rests largely on mammalian/*Drosophila* orthologues and on reviews, and there is a genuine scientific tension: the LysRS–Hint1–MITF "second-messenger" model vs. the *E. coli* "damage metabolite" reassessment. Both agree that an Ap4A hydrolase is needed to keep Ap4A low; they differ on whether the signalling output is physiologically deployed. NDX-4's catabolic role is robust under either interpretation.

Limitations and Knowledge Gaps

- 1. No direct *in vivo* functional study of *ndx-4* in worms.** There is, to our knowledge, no published *C. elegans* loss-of-function (RNAi/mutant) phenotype specifically characterising NDX-4's organismal role. The physiological narrative is inferred from biochemistry plus orthologue data.
- 2. Localisation is inferred, not measured, for NDX-4.** The "nuclear" assignment comes from the *Drosophila* Apf-EGFP orthologue (PMID: [17344088](#)) and mammalian work; the worm protein's subcellular distribution has not been directly reported.
- 3. Second-messenger vs damage-metabolite uncertainty.** Whether Ap4A acts as a deployed signal in worms (as proposed for mammalian mast cells) or is chiefly a toxic by-product kept low constitutively (PMID: [28516732](#)) is unresolved for *C. elegans*.
- 4. PRPP activity significance unknown *in vivo*.** The PRPP-pyrophosphatase side activity is demonstrated *in vitro* (PMID: [12370170](#)) but its physiological relevance (e.g., generating the glycolytic activator ribose-1,5-bisphosphate) is unestablished for NDX-4.

5. **Full in vivo substrate range not mapped.** The enzyme accepts various Ap4N substrates in vitro, but the dominant physiological substrate(s) and the actual Ap3A:Ap4A ratios it sets in worm tissues are not quantified.
 6. **Metal-cofactor profile of the worm enzyme** is not explicitly summarised here; Nudix enzymes generally require Mg^{2+}/Mn^{2+} , and orthologue data show Zn^{2+}/Mg^{2+} modulation, but NDX-4's precise cofactor dependence would benefit from direct citation.
-

Proposed Follow-up Experiments / Actions

1. **Direct localisation of NDX-4 in *C. elegans*.** Generate an endogenous *ndx-4::gfp* translational reporter (CRISPR knock-in) to test the predicted nuclear/chromatin-associated localisation and examine tissue-specific expression.
 2. **Loss-of-function phenotyping.** Create *ndx-4* deletion/RNAi strains and measure (a) steady-state Ap4A and Ap3A levels by LC-MS, (b) stress sensitivity (oxidative, heat, immune challenge), and (c) lifespan/reproduction, to connect the biochemistry to organismal physiology.
 3. **In vivo metabolite flux.** Use targeted metabolomics under baseline and stress conditions in wild-type vs *ndx-4* mutants to directly test the "off-switch" model and quantify the Ap3A:Ap4A ratio NDX-4 maintains.
 4. **Structure-guided catalytic-dead rescue.** Re-express catalytically dead NDX-4 (e.g., E56Q) in a null background to distinguish catalytic from any non-catalytic (scaffolding) roles.
 5. **Test the signalling axis in worms.** Probe for a LysRS/Hint1-equivalent pathway in *C. elegans* and whether Ap4A manipulation (via NDX-4 dosage) alters transcription of candidate target genes.
 6. **Assess the PRPP side-reaction in vivo.** Measure ribose-1,5-bisphosphate and nucleotide-biosynthesis flux in *ndx-4* mutants to determine whether the PRPP-pyrophosphatase activity has any physiological consequence.
 7. **Cofactor and inhibitor profiling.** Determine the metal requirement of recombinant NDX-4 and evaluate the asymmetric-Ap4A-hydrolase inhibitors identified in high-throughput screening ([PMID: 39521360](#), [PMID: 12697025](#)) as chemical-biology tools for worm studies.
-

Conclusion

C. elegans **NDX-4 (Q9U2M7)** is a Nudix-fold, metal-dependent **asymmetrical Ap4A hydrolase (EC 3.6.1.17)** that cleaves diadenosine tetraphosphate — and related dinucleoside tetraphosphates, but not triphosphates — at the fourth phosphate to yield **ATP + AMP**, via conserved catalytic glutamates (Glu52/56/103; $k_{cat} \approx 23 \text{ s}^{-1}$, $K_m \approx 8.8 \text{ }\mu\text{M}$), with a minor same-active-site PRPP-pyrophosphatase side activity. Its primary biological role is to keep intracellular Ap4A low and set the Ap3A:Ap4A ratio, functioning as the **catabolic off-switch of the LysRS-generated Ap4A stress signal** that governs concentration-dependent cell-fate decisions. It is a soluble intracellular enzyme with a probable nuclear site of action, and its crystal structure is the defining prototype of the asymmetrical Ap4A-hydrolase subfamily.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery